CPE464 – Computer Networks

Cisco Lab #1 – Looking at your Switch

Name: _____ Lab Section (please circle): 9am Noon 3pm

*This lab must be done with your partner. Please work together as a team. When working with other workgroups, do **not** tear your workstation down until all the workgroups you are working with have collected their data. If you have questions during the lab or are unsure about the instructions, please ask for help! It's fun to learn and we enjoy helping in the process.*

NOTE – there are two places on the worksheet that needs to be initialed, or you will NOT receive credit for this lab.

What you need from Canvas for this lab:

- Bring a copy of this lab procedure document
- Bring a copy of the lab 1 worksheet
- If you are not familiar with configuring IP addresses on Linux you should bring a copy of the IP Configuration document.

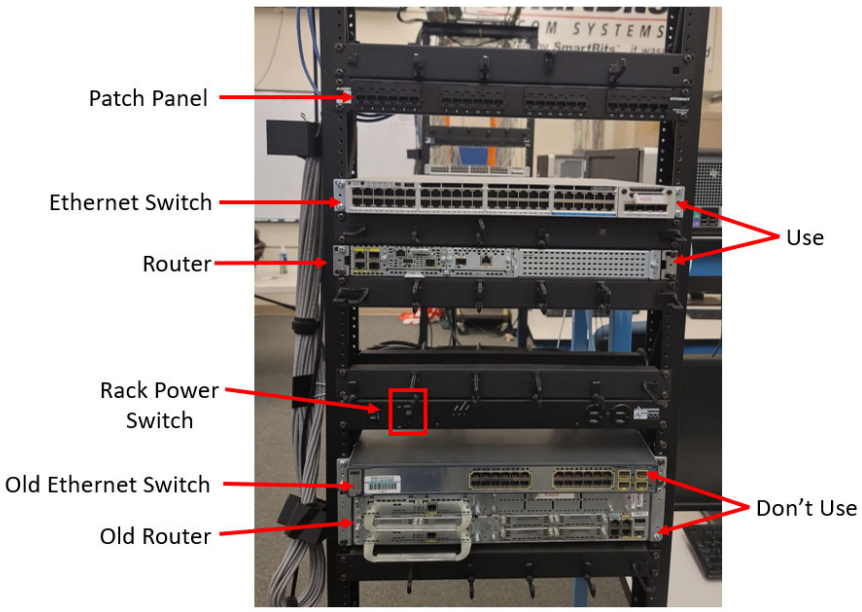
You need to have this lab worksheet and this document reviewed by the TA before leaving lab.

The purpose of this lab is to familiarize you with some of the equipment in the lab. You will also have a chance to experiment with different commands that are useful in configuring and understanding computer networks. This lab has three parts:

1) Configure your workgroup PCs and connect to the internet

2) Experiment with your Ethernet Switch
   a. Mac address table
   b. Port Spanning
   c. Spanning tree algorithm

Here is what the networking rack looks like at each workbench:



1. Initial Lab setup (with Internet Access)

A. Configure your workgroup PCs to talk on the Internet

In this step you are going to configure 2 PCs with your Ethernet switch (layer 2 device) and your router (layer 3 device). You are doing this just to see how the equipment works.

The lab workstations only run linux. If you are not familiar with networking in linux, see the IP Configuration document on polylearn.

You will need the following items for this section of the lab:

1 crossover cable (orange/yellow cable)
2 long straight-through cables (gray cable)
1 short straight-through cable

a. Boot your PC into Linux
b. Use the long straight-through cables to connect each computer's Ethernet port to a port on your switch.
c. Use the crossover cable to connect the switch to port 24 on the patch panel (this port connects to the real world e.g. the Internet).

Note – at this point we have not configured the addressing information for your computers – so they may not be able to use the network yet.

Configuring Addressing information

You will now need to setup your PC's IP address. To do this:

1) Follow the directions in the separate "Configuring IP Addresses" document
- Configure the PCs' IP address. The IP address for the computer can be found on a label near your workbench name or in the desktop background once the computer has booted. It should be in the range of 208.94.60.1 – 208.94.60.250
- Use the subnet mask 255.255.255.0
- Configure your default gateway to 208.94.60.254.
- Configure your DNS server addresses to 8.8.8.8 and 8.8.4.4.[1]
- Verify your IP settings. ping the other PC in your workgroup, ping an external site (e.g. www.calpoly.edu).
  - If any of your pings fail, stop and fix this before going on.

We will cover DNS later in this course. DNS translates between a name (e.g. www.calpoly.edu) and an IP address (e.g. 104.18.5.160)

Verify that you can Ping your partners PC (e.g. ping their IP address)

To check your PC's IP configuration, open a command prompt and enter the command *ifconfig*. Verify that your IP address and gateway. (Enter *ifconfig –h* to list all the options for ifconfig.)

Verify that your DNS settings are correct use the linux *dig* command on your PC (e.g. dig www.calpoly.edu). This should return IP address information a URL (e.g. for www.calpoly.edu).

Verify that you can talk to devices over the Internet using the ping command on your PC.

DO NOT continue until you can ping your partners computer and an outside site like Google or Cal Poly.

[1] These are google's public DNS servers.

Network diagrams are a great tool to help find and debug problems. While this network diagram is simple and does not tell us much, think about how hard it would be to debug a problem with a network that spans continents and time zones. Without some type of diagram, it would be very difficult to find and fix problems. For this purpose, companies maintain network diagrams.

Fill in the information in the network diagram in the worksheet including links and addresses (e.g. add IP addressing information).

Show the TA that you can ping both computers and an Internet site and have the TA initial your network diagram.

2. Looking at the Ethernet Switch's MAC address table

The reminder of the lab will look at the Ethernet Switch.

The MAC address table is used by the switch to map a device's Ethernet Address (e.g a PC's NIC address, also called a MAC address or a hardware address) to the physical outgoing port on the switch. This table allows the switch to forward Ethernet frames directly to a destination MAC address (so directly to a particular PC) and not have to broadcast the frame out all the switch's physical interfaces. Looking up the frame's Destination MAC address (layer 2 address) in the MAC address table and forwarding the frame out the correct interface is called switching and is why an Ethernet switch is called a switch.

MAC Address Table (found on Ethernet Switches)

MAC Address	Switch Interface
02:A3:BC:4A:11:22	GigEthernet 0/0/1

A. Rewiring your PC and Switch:

- Connect your left computer to the switch's port 13 using a straight through cable
- Connect your right computer to the switch's port 15 using a straight through cable
- Remove the connection from your switch to patch panel port 24 (at this point the switch should not be connected to the patch panel).

Since we will not be using the Internet for this part of the lab, remove both PC's default gateway:

- On each PC use the Linux command: *ip route del default*

B. Connecting to your switch command line interface via the Console Port

The Cisco switches in the lab are managed switches. These switches allow you to change configuration options of the switch via the switch's console access.

Wire to console port: To access the switch's administrative functionality, connect one of your PC's to your switch's console port on the back of y with the flat blue (or black) terminal cable. There should already be a blue/black cable attached to each PC. Use the top console port on the back left of the switch (do not use the bottom port – this will not provide you with full administrative access to the switch.

Run minicom: On the PC attached to the switch's console port, run the *minicom* (sudo minicom) application and then power up your power up your switch (if it does not already have power). Note the Cisco switch may take a few minutes to boot up (yes, minutes).

Once the switch has booted, using minicom, hit a return and you should see the switch's prompt.

Enter configuration mode on the switch: To configure the switch you must "enable" administrator access using the "enable" command.

```
SwitchX# enable
```

Press the Return key. You should see a password prompt:

```
Password:
```

The password is **`calpoly`**.

C. Viewing the MAC address table

Your switch maintains a MAC address table in order to know the link a device is attached to. The MAC address table maps the Ethernet MAC address of a machine to the interface that device is attached to on the switch.

The switch fills in in this table using a process called Address Learning. Every time a device (e.g. PC or router) sends an Ethernet frame through the switch, the switch will add the source MAC address from the Ethernet frame and the physical interface number of the port on the switch the frame arrived on into the MAC address table.

Your switch's MAC address table may be empty (note that both Windows and Linux devices are very "noisy" and periodically generate traffic even if the user is not doing anything). **To make sure some MAC addresses are in the MAC address table you can ping from one computer to the other computer. This ping will generate a number of frames that will allow the switch to learn about the PCs**. (If you cannot ping between your two computers, STOP and fix this problem!)[2]

From the switch console, enter the following commands to display its MAC address table:

```
SwitchX# show mac address-table
```

Write down your MAC address table below (dynamic address only). NOTE: To display only dynamic addresses, use the command **`show mac address-table dynamic`**.

Destination Address Destination Port

[2] ping is your friend. It helps you debug your networks. When wiring a network, always check that it works using ping. Its like putting print statements in your code. You are trying to figure out what works and what does not.

3. Connecting to other workgroups (Phase I wiring)

In this section we are going to look at the effects of the Address Learning protocol (and the MAC address table) on network traffic. Specifically, you are going to sniff traffic to determine the effects of using a switch (versus for example, a broadcasting hub).

- Use a straight-through cable to connect port 9 of your switch to a port on the patch panel (make sure you use a patch panel port that isn't occupied on the main panel at the front of the lab room).

- Using the patch panel in the front of the room, connect to another workgroup.

Make sure you can ping everyone in your partner workgroup.

Fill in the network diagram on the worksheet (titled "Connecting to other workgroups (Phase I), 4 PCs using Switches").

We will now use Wireshark (a packet sniffer) to capture and analyze your network traffic. We will do this from the **left** PC in your workgroup. For the rest of this lab, it is important that you always *sniff* from the **left** computer and *ping* from the **right** PC. Note that pinging from the wrong device or at the wrong time will produce the wrong results, so please follow the directions carefully!

On your workgroup's **left** PC:

- sudo wireshark in terminal window
- In Wireshark (aka packet analyzer):

    1. In the field labeled Filter, type: arp or icmp
    2. Select options from the Capture menu
    3. Select the only interface
    4. Select Promiscuous Mode
    5. Select Update List of Packets in Real Time
    6. Select Automatic scrolling in live capture
    7. Deselect all name resolution options
    8. Click Start

Next, you will be sending 50 pings from your right computer to the right computer of your partner workgroup.

Before you start your ping, how many total packets do you think you will see in Wireshark?

Here is the process to use (in this order!!) for sending the 50 pings:

1. Start wireshark on the left computer
2. Clear the arp cache on both workgroup's right computer
    - arp -n (this will list the arp cache)
    - arp -d <ip address to delete> (this will remove one entry from the cache, you must remove each entry manually, there is not way to just flush the cache)[3]
3. Clear the MAC address table on the Switch using:

    ```
    SwitchX# clear mac address-table dynamic
    ```

4. Start pinging (50 pings) from your right computer to your partner's workgroup's right computer

After doing the steps above to send 50 pings, how many ICMP packets are indicated on the left computer's analyzer[4]?

[3] In newer Linux versions you might have the 'ip' command that allows for flushing the entire ARP cache. You can try: sudo ip -s -s neigh flush all

[4] Note – your computers are pretty talkative and will send out different types of traffic on their own. Don't be surprised if you see some other random traffic in your packet traces. The arp and icmp filters in Wireshark should help reduce this problem.

Record the MAC address table on your switch.

Destination Address	Destination Port

What is the MAC address of the station you are pinging?

1. What port on your switch is associated with the MAC address of the computer you were pinging?

2. Why didn't the network analyzer see all the packets?

Clear the MAC address table again, and ping an address that is not on the network. This should generate broadcast traffic. Stop the Wireshark capture and start a new one. Use the same capture options as before.

Send 10 pings to the address 208.94.60.211 (ping from the right computer – this IP address is not on your network).

How many packets are indicated on the analyzer on the left?

Why don't you not see any icmp packets?

What IP address are the ARP packets trying to find MAC address information about?

4. Configuring the Switch Port Analyzer (SPAN) port

For this section, keep the same physical network configuration including your partner workgroup.

Because of the switch's address learning functionality, the switch can make it difficult to monitor traffic (e.g. using Wireshark) on your network. Initially your switch will broadcast out any frame it receives. Once the destination MAC address is learned by the switch the frames are only sent out on the link specified in the MAC address table.

To allow someone to monitor all of the traffic going through (in and out traffic) one of the switch's interface you will use the switch's SPAN feature. This feature allows traffic on one interface to always be duplicated and forwarded out another interface (the interface you configure for monitoring traffic). The original traffic is still forwarded out according to the address learning table but the frames are duplicated and set out on the monitoring interface.[5]

We are now going to use the SPAN feature of the switch to allow one port on the switch to monitor another port's traffic. You will configure the switch to allow the left computer to monitor the traffic to and from the right computer.

Enter the following commands on your switch:

[5] This is why using a hub works well for sniffing traffic. The hub automatically broadcasts out frames on all interfaces.

```
SwitchX# config term
SwitchX(config)# monitor session 1 source int gi1/0/2 both[6]
SwitchX(config)# monitor session 1 destination int gi1/0/1
SwitchX(config)# exit
SwitchX# show monitor session 1
```

Reset Wireshark on the left computer and begin sniffing. On the right computer, ping your partner workgroup's right computer. Verify that you see all the packets in Wireshark on the left PC.

Why is port monitoring important in a switched environment?

To turn this feature off, enter the following:

```
SwitchX# config term
SwitchX(config)# no monitor session 1
SwitchX(config)# exit

SwitchX# show monitor session 1
```
(should now show nothing.)

5. Handling redundant links – The Spanning Tree Protocol (STP)

In this section you will configure the switch's spanning tree protocol. This protocol allows the switch to create a logical tree removing any loops in your physical network. Remember that one of the benefits of a tree structure is that it contains no loops. Once an Ethernet frame is put into the Ethernet network, it will continue to be forwarded until it reaches the destination or the end of the network. If your Ethernet network has a loop any frame sent into the network will continue being forwarded until it takes down your network. The Spanning Tree Protocol takes your arbitrary physical topology and makes it work as a logical tree topology.

One negative of the STP is that it does not consider traffic patterns. It may create a logical tree that requires traffic to follow paths much longer than would be necessary in your original physical network. This means if you purposefully wire up a network to carry traffic efficiently, the SPT may not use your intended topology and may require the network traffic to use non-optimal paths.

One nice thing about the STP is that if a link is taken down (e.g. broken) the STP will run again and bring back up links to reconnect the network. Assuming there are redundant physical links, your network can recover from a broken connection without human intervention.

This section will demonstrate the **Spanning Tree Protocol** features of the switch. Make a *second* connection from your switch to your partner workgroup's switch, using **port 8 on both switches**. Now, you should have switch ports 8 *and* 9 connected to each other via the patch panel.

Now we need to turn off the Spanning Tree Protocol feature on both switches.

It is recommended to close Wireshark at this point, however if you want to see the traffic that is generated by the switch you can leave it on. **Do so at your own risk… it might lockup the PC for a little while.**

```
SwitchX# config term
SwitchX(config)# no spanning-tree vlan 1
```

[6] The 'gi' in this command is for the gigabit ethernet (1000 megabits/second) interface.

```
SwitchX(config)# end
```

Send 50 pings to your neighbor's right computer.

1. After sending the 50 pings (or just waiting a while) which ports are active (light for port is green)?

2. How many of the pings succeed?

3. What is causing the pings to fail?

Disconnect port 9 on the switch to break the loop. Be sure all pings have stopped before proceeding.

Enter the following commands to enable Spanning Tree on the switch:

```
SwitchX# configure terminal
SwitchX(config)# spanning-tree vlan 1 (note - this is the command that turns on the STP)
SwitchX(config)# end
```

What is the status of the ports (look at the switch leds – yellow is down, green is up)?

How long before the ports are active/down (so how long to stabilize)?

From your right computer, ping your partner workgroup's right computer.

Record the Mac Address Table on your switch.

Destination Address	Destination Port

Reconnect port 9 to re-establish the redundant path to the other switch.

How long until this port becomes active?

Compare your answer to the above question to your partner workgroup's answer.

From your right computer, ping your partner workgroup's right computer.

Record the Mac Address Table on your switch.

Destination Address	Destination Port

Which path is being used between the two switches? (port to port)

Enter the following command to view the spanning tree state on port X:

```
SwitchX# show spanning-tree interface gigabitethernet1/0/X
```

Use this command for your switch ports 8 and 9.

4. Is port 8 FORWARDING or BLOCKING?

5. Is port 9 FORWARDING or BLOCKING?

Compare your answers to the above questions to your partner workgroup's answers.

Now send 50 pings to your neighbor's right computer.

Have we fixed our problem from before? How many of the pings succeed?

A. Testing Fault Tolerance using STP

The nice thing about using the STP protocol is that it can automatically recover from network problems. If a physical connection between two switches that is part of the Spanning Tree is broken (goes down for some reason) the STP protocol will generate a new spanning tree taking into consideration all of the remaining physical links.

Make the sound of a backhoe (or a ship dropping anchor) while you disconnect port 8 from your switch.

How long until the blocked port becomes a forwarding port?

What is the status of port 8?

Reset the MAC address table, and from your right computer, ping your partner workgroup's right computer.

Record the MAC address table on your switch.

Destination Address	Destination Port

Which path is now being used the two switches? (port to port)

6. When done:

1) Turn off the power to your network rack
2) Return the cables and clean up your workgroup
3) Have the TA check your lab and worksheet to see that you completed the work. They will check you off before you leave.